

Alejandro Jaimes

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

Orlando, FL

May 2028

EXPERIENCE

Software Engineer Intern

May 2026 - Present

Institute of Simulation and Training

Orlando, FL

- Constructed a real-time spatial anchoring system using **AR Foundation**, enabling **8** users within **5cm** accuracy.
- Developed networked player-tracker with **Photon**, matching data for **XR** devices at low latency for live demos.
- Implemented runtime **XR subsystem** detection for hardware-aware kiosk mode, reducing demo setup by **100%**.
- Designed backward-compatible multiplayer integration system, maintaining **100%** single-player functionality while adding new multiplayer features across **2** distinct codepaths.

President

Apr 2026 - Present

Graphics Programming Knights

Orlando, FL

- Established Graphics Programming Knights, a non-profit student organization, growing membership to **165+**.
- Organized the first community-wide Render Jam competition resulting in **40+** participants and **5** projects.
- Orchestrated **2 workshops per term** on pathways into graphics programming, covering essential mathematics, game development engines, and GPU programming with student attendance of **20+**.
- Expanded club reach by **40%**, establishing partnerships to have workshops and hackathons with **3** organizations.

Outreach Board

Jan 2026 - Present

Knight Hacks

Orlando, FL

- Boosted online community engagement online with strategic content creation and event awareness by **57%**.
- Increased Knight Hacks' activity averaging **5k+** views on Instagram Reels by editing and participating in videos.
- Expanded event attendance by **15%** compared to previous semesters through targeted outreach campaigns on social media and speaking to the public, increasing participation with an average of **40+** people per meeting.

PROJECTS

sFun (SNES emulator) | C++, C, CMake, Assembly, SDL2

- Designed and implemented a **65c816 CPU** emulator supporting, dynamic register width handling using **C++**.
- Built an extensible opcode dispatch system using function pointer tables and implemented core CPU instructions (LDA, REP/SEP, branching, stack operations, JSR/RTS) with correct flag behavior.
- Developed a memory subsystem for SNES RAM/ROM access, enabling instruction fetch, decoding, and execution.
- Engineered robust control flow correctness for relative branching, subroutine calls, and stack-based return handling ensuring **100%** correctness on test cases.

Next Step | React, Tailwind, Next.js, TypeScript, Python, Flask, Vapi

- 3rd place winner** of State Farm's Sponsor Challenge at Shell Hacks 2025 out of **50** projects submitted.
- Created an AI voice agent delivering interactive with personalized feedback across **3** different insurance categories.
- Implemented backend services to process and structure conversational data generating dynamic agent responses, reducing response time by **66%** using **Python**, integrated with the **Vapi API**.
- Created **7** interactive components for optimal experience with **TypeScript**, **Next.js**, and **React** for frontend.

TK's Revenge | LuaU, Roblox Engine

- 2nd Place winner** at Knight Hacks' Project Launch competition out of **35** projects and **150+** participants.
- Winner of Public Vote** with over **90+** votes, average playthrough of **11** minutes and approval rating of **100%**.
- Implemented **AI** pathfinding in **LuaU** enabling **3** enemies to dynamically engage with players simultaneously.
- Created different movement techniques for **2** different classes to use, ensuring diversified range of options.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, HTML, CSS, JavaScript, TypeScript, Assembly, Java

Frameworks/Libraries: Unity, Next.js, Node.js, Tailwind CSS, React, Flask, Unreal Engine, Three.js, OpenCV, Mediapipe, Vapi, SLD2, OpenGL, CUDA, CMake